

AUDIO SLIDERS: MEASURING WHAT A SLIDER DOES TO MUSIC

Anonymous Authors
Anonymous Affiliations
anonymous@ismir.net

ABSTRACT

A slider is a low-rank update to a frozen generative model whose strength is set at inference time, so one prompt and seed can be moved along a single axis. Sliders for music are scored by text–audio embedding similarity, on axes someone chose by writing a prompt pair. We measure what that leaves out, on Stable Audio Open and ACE-Step: whether a waveform property follows the slider, whether two unrelated quality predictors still rate the result as music, how much of the piece survives, and what else moves. A trained slider matches prompt-pair guidance with a third of the forward passes, and prompt interpolation fails. Two sliders pass the embedding test and fail the waveform test. We then take axes from real music: independent components of embeddings of 14,985 recordings include arousal and valence, and the model’s output spans 28–83% of the real spread along them. Sliders along these axes, trained from the axis’s own tags or with no text from the model’s clips, follow them on held-out prompts (rank correlation up to 0.91) and carry a clip across the real-music mean, 2.4 real-music standard deviations inside the usable span. Principal axes of the generator’s own activations describe production instead.

1. INTRODUCTION

A text-to-music model takes a prompt and returns a clip. A musician who likes the clip but wants it a little happier, or fuller, or with richer harmony, has to rewrite the prompt and accept a different piece. A slider offers something else: keep the prompt and the seed, and move the result along one axis by a chosen amount.

Sliders exist for image diffusion as low-rank adapters with a runtime scale [1], and the idea of discovering the axes themselves comes from SliderSpace [2]. Both have reached audio models as baselines [3, 4]. In that work a control counts as working when the output moves toward the attribute text in a joint text–audio embedding and stays close to the original.

For music that leaves four things open. The embedding can move while the property a listener would name does not. A control can reach its target by wrecking the clip,

and similarity to the original does not say whether what remains is music. Nothing in the score says what else moved. And the axes are chosen by whoever writes the prompt pair, with no evidence that they are the directions along which music varies or that the model can reach their ends.

This paper measures those things on two open backbones. Its contributions:

- A protocol that scores a control on a waveform descriptor fixed in advance, in two unrelated embedding models, on two unrelated quality predictors anchored by real recordings, on similarity to the unsteered piece, and on what else moved. Effect is reported only over the *usable span*, the positions at which quality has not dropped.
- A comparison of four ways to move the same attribute under that protocol. A trained slider reproduces prompt-pair guidance with one forward pass per step instead of three, activation steering is competitive with both, and interpolating the text conditioning fails. Two sliders pass the embedding test and fail the waveform test.
- Axes taken from real music. Independent components of embeddings of 14,985 recordings include arousal and valence. Projected on these axes, ACE-Step’s output over 648 prompts spans 28–83% of the spread of real recordings, and Stable Audio Open shows the same pattern.
- Sliders along those axes, trained two ways: from a prompt pair built from the axis’s own tags, and with no text from two sets of the model’s clips sorted along the axis. Both follow the axis; the first moves further, the second keeps more of the piece and is more selective. We also report what did not work: training between sets of real recordings, and principal axes of the generator’s own activations, which turn out to describe production rather than musical content.

2. RELATED WORK

Sliders. Concept Sliders [1] train a LoRA [5] whose output matches the frozen model’s prediction shifted along the difference between a positive-prompt and a negative-prompt prediction. SliderSpace [2] removes the prompt pair: it takes principal components of CLIP embeddings of many images of one concept and trains one adapter per



component. Measured Sliders [6] define image sliders by differentiable measurements and hold other measurements still; we adopt their selectivity score.

Steering audio models. FreeSliders [3] apply the Concept Sliders target at inference time without training, on image, video, and audio models; most of their audio concepts are sound effects. TADA [4] steers cross-attention activations of audio diffusion transformers, with steering vectors from prompt pairs or from sparse-autoencoder features selected with prompt pairs. Facchiano et al. [7] compute difference-in-means vectors for tempo and timbre, AnchorSteer [8] trains text-free concept modules on self-generated clips, and Singh et al. [9] find sparse features in autoregressive music models. These score steering by text–audio alignment and perceptual distance to the unsteered output. TADA checks two concepts against a descriptor, and Wang [10] shows with a key estimator and a beat tracker that apparent agreement with an instruction can be the model’s prior. None scores every control against a waveform property fixed in advance, compares against signal processing, or reports what else moves.

Where axes come from. In image diffusion, principal components of a U-Net’s bottleneck activations are semantic [11, 12]. Guinot et al. [13] learn sparse features of music embeddings to steer retrieval, and independent components of text embeddings are known to be more interpretable than principal ones [14]. We found no work that takes axes from embeddings of real music and trains generator controls along them.

Diversity against real music. Recall and coverage in an embedding space give one number per model [15]. Slendebroek and Metaxa [16] compare two commercial systems with human recordings on hand-built signal features and report homogenisation. Our comparison is per axis, on axes learned from real recordings, for open models.

Effects with known parameters. RIME [17] builds paired edit data by applying effects with known parameters. We use such effects as a baseline that a learned control has to justify itself against.

3. METHOD

Backbones. Two frozen text-to-music models sit behind one interface. Stable Audio Open 1.0 [18] is a latent diffusion transformer with a v -prediction objective, sampled with DPM-Solver++(2M) for 50 steps at guidance 7. ACE-Step 1.5 XL turbo [19] is a rectified-flow transformer, $x_t = (1-t)x_0 + t\epsilon$ with the network predicting $\epsilon - x_0$, distilled to 8 Euler steps without guidance. Everything below is written for a generic prediction $f_\theta(z_t, t, c)$.

A slider. A slider is a set of low-rank updates, one per linear layer in the attention and feed-forward blocks. With position $s \in \mathbb{R}$ a layer computes

$$y = Wx + s \frac{\alpha}{r} BAx, \quad (1)$$

with $r = 4$ and $\alpha = 1$. s is read at every forward pass and can differ per sample. Sliders add, and a slider can be switched on only for $t \leq t_0$ (gating).

Training from a prompt pair. Let c^+ and c^- be a base prompt c with a phrase for each end of the attribute appended. Following Concept Sliders, for $s \in \{+1, -1\}$ the slider is trained so that

$$f_{\theta,s}(z_t, t, c) \approx f_\theta(z_t, t, c) + s \eta (f_\theta(z_t, t, c^+) - f_\theta(z_t, t, c^-)), \quad (2)$$

with the right-hand side from the frozen model. z_t comes from sampling for a random number of steps with the slider active. We use $\eta = 4$ for Stable Audio Open and $\eta = 2$ for ACE-Step, 1,000 iterations, and 48 training prompts disjoint from the evaluation prompts.

Training from two sets of clips. The second trainer uses no text. Given sets H and L of latents, we minimise the model’s denoising loss with the slider at $+1$ on H and -1 on L :

$$\mathbb{E}_{x_0 \in H} \|f_{\theta,+1}(z_t, t, c) - u\|^2 + \mathbb{E}_{x_0 \in L} \|f_{\theta,-1}(z_t, t, c) - u\|^2, \quad (3)$$

where u is the regression target for (x_0, ϵ, t) . One update serves both ends with opposite sign, so to first order what the sets share cancels. H and L are the top and bottom 20–30% of the model’s own clips along some per-clip measurement, taken *within each prompt* so that prompts are balanced. The measurement can be a descriptor, a quality score, or the projection on a direction in an embedding space, and it can be residualised against other measurements first. No differentiable path through the decoder or the embedding model is needed.

Axes. We take directions from three places. (i) Principal components of CLAP embeddings of 1,024 clips of one concept, as in SliderSpace. (ii) Principal and independent components [14] of CLAP [20] and MuQ-MuLan [21] embeddings of 14,985 real recordings from FMA [22], after removing the 40% most vocal clips and each genre’s mean. (iii) Principal components of the generator’s own activations: the mean cross-attention output of every block for 768 clips, with each prompt’s mean removed.

Other ways to move an attribute. Under identical prompts, seeds, and sampler we also run the right-hand side of Eq. (2) directly at every step (the training-free recipe of FreeSliders, three forward passes per step); activation steering, which adds the mean difference in cross-attention output between c^+ and c^- to every block [4]; interpolation of the text conditioning toward c^+ or c^- ; and, where one exists, a conventional effect applied to the unsteered clip.

4. MEASUREMENT

For each prompt and seed the same initial noise is rendered at every slider position; we call this a trajectory. All numbers use 24 prompts held out from training, three seeds, and ten-second clips unless stated.

Does a measured property follow the slider? Each slider is paired in advance with a descriptor computed from

the waveform: spectral centroid, onset rate, beat-tracked tempo [23], energy decay time, percussive energy share, low-band and stereo-side energy ratios, key clarity, major-minus-minor key fit, harmonic change rate. We report the mean Spearman correlation between position and descriptor over trajectories (ρ) and the change in units of the descriptor’s standard deviation across unsteered clips.

Does a second model agree? The projection of each clip on the slider’s text direction is computed in CLAP and again in MuQ-MuLan, which shares no weights or training data with it.

Does it still sound like music? Every clip is scored by Audiobox Aesthetics [24] (content enjoyment, CE, 1–10) and by SongEval [25] (musicality, 1–5), a head on MuQ features trained on human ratings of generated songs. Both are anchored by 2,000 excerpts of real recordings. The *usable span* is the set of positions, walking outward from zero, over which mean CE stays within 0.5 of the unsteered clips. Effect sizes are reported inside it.

How much of the piece survives? CLAP similarity between each clip and the same trajectory at position zero.

What else moves? The slope of every descriptor against position, in standard deviations per unit: a leakage matrix. A tag panel of 80 instrument, genre, mood, and character words, scored in CLAP, puts the change in words.

Coverage. For a direction d found in real recordings, the coverage of a model is the standard deviation of its clips’ projections on d divided by that of the recordings. We report it over all prompts, and after removing each prompt’s mean from the generated clips and each genre’s mean from the recordings. A slider’s reach along d is reported in the same unit.

5. RESULTS

Prompt-pair sliders. Content enjoyment averages 6.12 over 2,000 real recordings, 6.95 for unsteered ACE-Step clips and 6.04 for Stable Audio Open. Table 1 and Figure 1 report eight ACE-Step sliders over nine positions. Where a descriptor was assigned in advance it rises with the slider, and MuQ-MuLan agrees with CLAP on the direction for all eight. Enjoyment stays above the real-music mean across the range for all but tension, whose usable span ends at +1. SongEval gives the same picture from different training data: unsteered clips score 2.77 in musicality against 2.78 for the real recordings, the usable span it implies is identical for six of the eight sliders and half a step wider for the other two, and the two predictors agree over clips with Spearman 0.52–0.67. Sliders trained on ten-second clips carry over to thirty seconds: brightness stays ordered on every trajectory ($\rho = 1.00$), ensemble on 92%, and unsteered enjoyment is 7.34.

Four ways to move the same attribute. Table 2 runs the alternatives on the same prompts and seeds. The trained slider reproduces prompt-pair guidance, its own training target, with one forward pass per step instead of three: it is higher on brightness and harmony and within the confidence interval elsewhere. Activation steering is within

0.05 of the slider on four of six descriptors and keeps more of the piece (0.83–0.87 against 0.74–0.84), while the slider moves its descriptor further on brightness, ensemble, and harmony. Interpolating the text conditioning has the lowest correlation on every attribute and breaks the audio at the ends.

The embedding says yes, the waveform says no. On Stable Audio Open, sliders trained from the prompt pairs for note density and percussion move the output along their CLAP direction at every setting, and neither moves its descriptor: the rank correlation with onset rate or percussive share never exceeds 0.37. Scored by embedding alone both would be reported as working. The descriptor has to be sound too: with a tempo estimate prone to octave errors the tempo slider scored $\rho = -0.01$, and with a beat tracker it scores 0.57 and moves the mean tempo from 90 to 167 BPM.

Gating. On the 50-step model a slider at ± 1 takes the piece with it. Switched on from step 21, the brightness slider keeps 1.41 of its 2.37 standard deviations of range, similarity to the unsteered clip rises from 0.67 to 0.85, and the loss in enjoyment falls from 1.0 to 0.3 points (Table 3).

What else moves, and cancelling it. The mood and melody sliders both raise the spectral centroid by 0.8 standard deviations per unit. Adding the brightness slider at -0.53 and -0.55 times their position brings the leak to -0.02 and $+0.09$, raises similarity to the unsteered clip at ± 1 from 0.85 to 0.90 and from 0.87 to 0.92, and lowers the CLAP direction correlation from 0.92 to 0.69 and from 0.93 to 0.72. Part of what the embedding scores as happier or more melodic is brightness.

Axes of real music. Fitted on two halves of the recordings and matched one to one, the leading eight axes agree with mean cosine 0.98 (MuQ-MuLan, principal), 0.94 (CLAP, principal), 0.90 (MuQ-MuLan, independent), and 0.83 (CLAP, independent). Of 1,024 sparse-autoencoder features on CLAP, only 9 are found by two seeds. The independent components of MuQ-MuLan read most like music: one runs from quiet, minor-key, and dreamy to aggressive, dry, and rhythmic, another from dark and distorted to playful and simple, the arousal and valence of the circumplex model [26]; others contrast jazz with electronic music, plucked and bowed strings with synthesizer and choir, and piano with organ and rock.

Coverage. Along every one of these axes the model’s output is narrower than real music (Figure 2). Over 648 prompts crossing genre, instrument, and mood, ACE-Step spans 28–83% of the real spread; seeds of one prompt span 38–69% of what recordings of one genre span. The average generated clip sits 1.35 standard deviations toward the quiet end of the arousal axis and 1.36 toward the playful end of the valence axis. On CLAP’s first three principal axes ACE-Step covers 35–39% and Stable Audio Open 48–52%. The top eight principal directions of the generated corpus contain 31% of the top eight of the real one.

Sliders along real-music axes. We turned the axes into sliders in two ways (Table 4). The set trainer takes the top

Slider	Descriptor	ρ	Ordered	MuQ ρ	Span	Moved	Kept	CE ₀	CE _{ends}
brightness	centroid_oct	0.97	1.00	0.80	-1.0 to +2.0	4.41	0.74	6.95	6.04
density	onset_rate	0.78	0.92	0.86	-0.5 to +2.0	1.25	0.84	6.95	6.09
mood	majorness	0.39	0.75	0.88	-1.0 to +2.0	0.87	0.79	6.95	6.65
harmony	harmonic_change	0.41	0.76	0.73	-1.5 to +2.0	0.84	0.83	6.95	6.64
groove	—	—	—	0.56	-1.5 to +2.0	—	0.79	6.95	6.88
ensemble	pc	0.60	0.90	0.77	-1.5 to +2.0	0.92	0.80	6.95	6.56
tension	—	—	—	0.86	-1.5 to +1.0	—	0.83	6.95	5.64
melody	key_clarity	0.48	0.78	0.89	-1.0 to +2.0	0.97	0.80	6.95	6.47

Table 1. Prompt-pair sliders on ACE-Step 1.5 XL turbo. ρ : mean rank correlation between position and descriptor. Ordered: share of trajectories with correctly ordered ends. MuQ ρ : the same correlation for the MuQ-MuLan direction score. Span: usable span. Moved: descriptor change over the span in standard deviations of unsteered clips. Kept: CLAP similarity to the unsteered clip at the ends of the span. CE: content enjoyment unsteered and at the extreme positions.

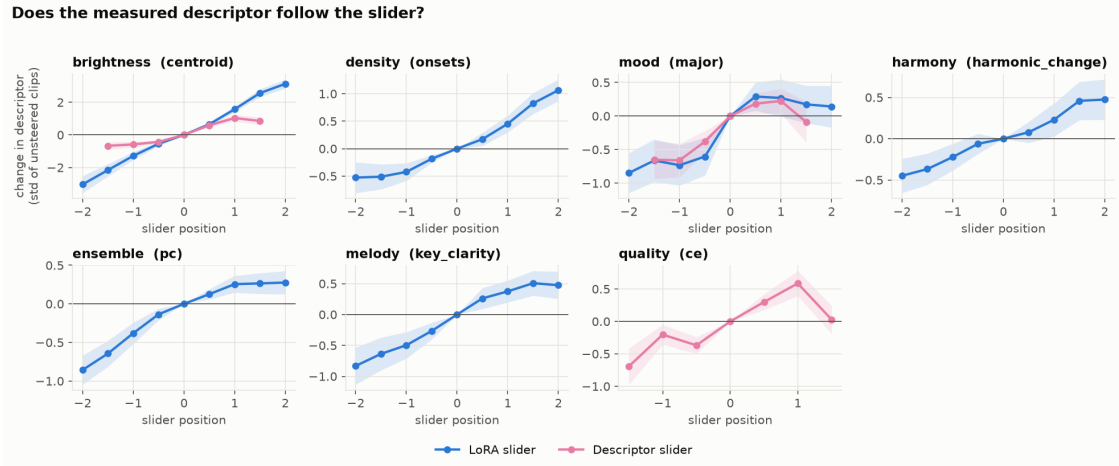


Figure 1. ACE-Step: change in each slider’s descriptor against position, in standard deviations of unsteered clips, for prompt-pair sliders and for sliders trained between two sets of clips.

Slider	Slider		Guidance		Act. steer.		Interp.	
	ρ	MuQ	ρ	MuQ	ρ	MuQ	ρ	MuQ
brightness	0.97	0.80	0.89	0.81	0.93	0.81	0.49	0.35
density	0.78	0.86	0.74	0.81	0.71	0.86	0.10	0.38
mood	0.39	0.88	0.45	0.87	0.41	0.88	0.25	0.63
harmony	0.41	0.73	0.08	0.68	0.24	0.72	-0.01	0.23
groove	—	0.56	—	0.71	—	0.58	—	0.12
ensemble	0.60	0.77	0.55	0.75	0.58	0.67	0.25	0.25
tension	—	0.86	—	0.88	—	0.91	—	0.69
melody	0.48	0.89	0.51	0.83	0.53	0.83	0.31	0.38
CE at ends	6.37		6.54		6.52		4.95	
Passes per step	1		3		1		1	

Table 2. Four ways to move the same attribute on ACE-Step, same prompts and seeds. ρ : rank correlation of position with the waveform descriptor (— where none was assigned); MuQ: with the MuQ-MuLan direction. CE at ends: mean content enjoyment at the two extreme positions, averaged over sliders.

and bottom 20% of 24 seeds per prompt along the axis. The second route reads the tags at the two ends of an axis and uses them as the prompt pair in Eq. (2); for the arousal axis these are “quiet, dreamy, melancholic, minor key” and “aggressive, energetic, rhythmic, dry”. Either way the output is scored by its projection on the axis from real recordings.

All twelve sliders follow their axis. The prompt pair is

On after	Moved	CLAP	Chroma	CE
0	2.37	0.67	0.68	5.15
10	2.11	0.71	0.71	5.34
14	1.98	0.74	0.74	5.45
21	1.41	0.85	0.84	5.87
29	0.42	0.97	0.97	6.08

Table 3. Timestep gating for the brightness slider on Stable Audio Open at positions ± 1 : descriptor moved (std), CLAP and chroma similarity to the unsteered clip, content enjoyment. Unsteered CE is 6.16.

the stronger route on four of five axes, with rank correlations of 0.78–0.91 and 1.6–1.9 standard deviations of real music between -1 and $+1$. Unsteered clips sit 1.21 standard deviations toward the quiet end of the arousal axis; the slider takes the same prompts and seeds across the real-music mean to -0.26 at position -2 , and to $+2.16$ at $+1$, where its usable span ends. The standard deviation of the model’s whole output over 648 prompts on this axis is 0.83. Set-trained sliders move less (0.44–1.19), keep more of the piece (0.84–0.87 against 0.73–0.85), and hold both quality scores level at both ends, and on valence they are the more reliable (93% of trajectories ordered against 75%).

Each trainer moves what it was trained on. Sorting clips

The model explores less of each musical axis than real music does

Axes: independent components of MuQ-MuLan embeddings of 14,985 FMA recordings. Model: ACE-Step 1.5 XL turbo, 15,552 clips.

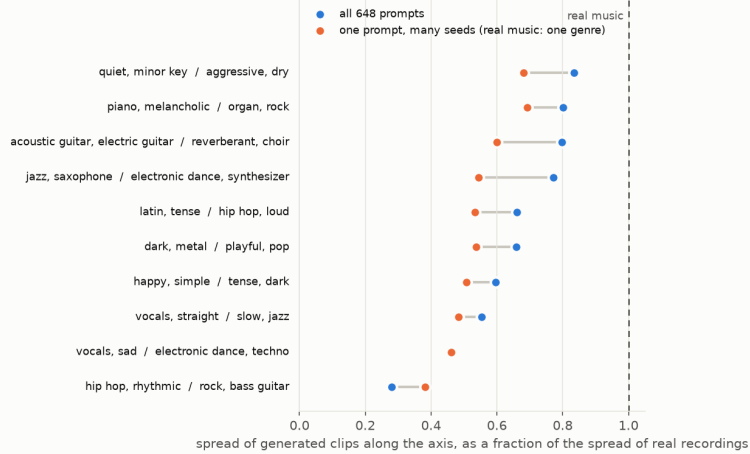


Figure 2. Coverage of ten real-music axes by 15,552 ACE-Step clips. Each axis is labelled by the tags at its two ends.

Axis	From	ρ	Ord.	Real	Seeds	Kept	CE	Mus.
acoustic/electr.	sets	0.55	0.93	0.64	1.48	0.86	7.11	2.68
arousal	pair	0.91	1.00	1.85	3.35	0.80	6.84	2.71
arousal	sets	0.82	0.99	1.19	2.14	0.84	7.13	2.70
classical/funk	sets	0.64	0.96	0.77	1.65	0.85	7.12	2.69
jazz/electronic	pair	0.86	0.97	1.86	3.78	0.82	7.06	2.72
jazz/electronic	sets	0.61	0.94	0.61	1.23	0.87	7.09	2.70
piano	pair	0.78	0.97	1.56	2.93	0.80	6.78	2.74
piano	sets	0.63	0.96	0.75	1.42	0.87	7.08	2.69
strings/synth	pair	0.80	0.97	1.78	3.28	0.73	6.70	2.64
strings/synth	sets	0.40	0.82	0.44	0.81	0.87	7.09	2.69
valence	pair	0.42	0.75	0.86	1.71	0.85	6.75	2.67
valence	sets	0.50	0.93	0.62	1.23	0.87	7.07	2.73

Table 4. Sliders along axes of real music on ACE-Step, positions -1 to $+1$. From: a prompt pair made of the axis’s tags, or two sets of the model’s clips. ρ , Ord.: rank correlation between position and the output’s projection on the axis, and share of trajectories with -1 and $+1$ in the right order. Real, Seeds: movement along the axis between -1 and $+1$ in standard deviations of real recordings and of one prompt’s seeds. Kept: CLAP similarity to the unsteered clip. CE, Mus.: enjoyment and SongEval musicality at ± 1 (unsteered: 6.95 and 2.77).

90% of the within-prompt variance of ACE-Step’s cross-attention outputs and reproduce across disjoint prompt sets (0.82). Steering along the first two raises level and lowers spectral flatness ($\rho = 0.77$ and -0.64) and moves tags from lo-fi toward complex and live. The other six leave at least 95% similarity to the unsteered clip at twelve natural standard deviations and act differently from prompt to prompt (consistency 0.10–0.21). Unlike the image case [12], the leading directions here describe production; arousal, valence, and instrument contrasts appeared only in embeddings of real music.

Training between sets of real recordings. With H and L drawn from the real recordings themselves, the slider damaged the clip equally in both directions. A penalty that ties the mean of its predictions at ± 1 to the frozen model removed the damage and left almost no steering: tempo did not move and major-minor fit moved from 0.09 to 0.15. Real music was useful for finding axes and the model’s own clips for training along them.

6. LIMITS

No listening test was run. Quality rests on two learned predictors and meaning on tags and signal descriptors, which are proxies: energy decay time is confounded by sustained instruments, and SongEval was trained on songs with vocals. Most clips are ten seconds long. The usable-span tolerance of 0.5 CE is a choice. Coverage depends on our prompt list and on FMA as the sample of real music, and both embedding models were trained on real recordings. Axis labels come from an 80-word vocabulary and a different vocabulary would name the same direction differently. The internal-axis result is for token-averaged cross-attention outputs of an eight-step distilled model.

by a waveform descriptor, residualised against loudness, brightness, and enjoyment, gives sliders that are selective. Over positions -1 to $+1$ the set-trained harmony slider reaches $\rho = 0.72$ on harmonic change rate with selectivity 6.6, against 0.27 and 0.8 for the prompt-pair slider; density and ensemble show 4.7 and 2.4 against 1.5 and 1.5. The embedding tells the opposite story: the MuQ-MuLan text direction for harmony follows the prompt-pair slider ($\rho = 0.63$) and ignores the set-trained one (-0.07). Neither kind of score can stand in for the other. Set-trained sliders stop at ± 1 : beyond it similarity falls below 0.72 and musicality drops by 0.3–0.5 in both directions while CE stays flat, a failure only the second predictor shows. Tempo did not become a slider this way ($\rho = 0.06$).

Axes inside the generator. Eight principal axes carry

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